

Bara Kharseh

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EDUCATION & QUALIFICATIONS

University of Toronto

September 2021 – April 2026

HBSc in Computer Science & Human Biology; Arts & Science Internship Program

Toronto, ON

- **Relevant Coursework:** Software Design, Introduction to AI, Introduction to ML, Introduction to Databases.
- GPA: **3.90/4.00** (Dean's List Scholar every year).

Languages & Technologies: Python, Java, JavaScript, HTML/CSS, SQL, Git, Django, React, Celery (distributed task queue), Docker.

EXPERIENCE

The Hospital for Sick Children (SickKids)

May 2024 - May 2025

Full-stack Software Developer Intern

Toronto, ON

- Contributed to the development of **OTTER** [🔗](#), an **AI-driven** research web application used by **40+** institutions across **10** countries, to classify tumour samples into one of **455** subtypes using gene expression data.
- Launched CAPYBARA, a web portal that streamlines the use of OTTER for pathologists at **SickKids, Mount Sinai Hospital, & Aga Khan University Hospital** in Nairobi, from scratch & scaled to **400 files in 8 months**, containerizing & deploying the application using Docker.

RESEARCH

Krembil Brain Institute | Toronto Western Hospital

August 2025 - April 2026

Neurophysiology Research Student

Toronto, ON

- **Project Title:** Beta oscillatory frequency shifts in the human subthalamic nucleus during the auditory oddball task in Parkinson's disease.
- Analyzed power spectral densities from **30+** Parkinson's deep brain stimulation procedure microelectrode recordings to investigate beta frequency shifts in the human subthalamic nucleus during a cognitive task.

The Hospital for Sick Children (SickKids)

September 2024 - May 2025

Cancer Genomics Research Student

Toronto, ON

- **Project Title:** Co-essentiality patterns reveal novel gene targets in childhood cancer.
- Led a first-author **13,313-sample** tumour analysis to address lack of childhood cancer therapies.

PROJECTS

SciRef [🔗](#) | *React, Flask, Tailwind CSS, JavaScript, HTML, CSS*

September 2023

- Demonstrated efficient collaboration within a team of **3** developers by implementing a full-stack web application that generates customizable scientific references given a URL to a scientific journal.
- Created a robust solution for scientific reference generation by employing **React & Flask** frameworks.

Doctor Clinic Management System [🔗](#) | *Java*

June 2022 – August 2022

- Developed a terminal application in a **team of 8** students with login functionality to manage doctor clinics
- Implemented **design patterns**, adhered to **Clean Architecture** & followed an **MVC architecture** to create a **clean & extensible codebase**
- Designed & implemented over **50** distinct commands enabling users to execute actions including signing in, scheduling appointments, & accessing prescription information

AWARDS

Dorothy Helen McRobb Scholarship

February 2026

Woodsworth College - University of Toronto

Toronto, ON

- Awarded to students who demonstrate academic merit.

Entrance Scholarship

September 2021

Woodsworth College - University of Toronto

Toronto, ON

- Awarded to exceptional incoming U of T students for academic & personal achievement.

Governor General's Academic Medal [🔗](#)

June 2021

L'Amoreaux Collegiate Institute

Toronto, ON

- Awarded to graduating Ontario secondary students with highest Grade 11 & 12 average in their school.